12.159

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Question

The determinant

$$\begin{vmatrix} 1+b & b & 1 \\ b & 1+b & 1 \\ 1 & 2b & 1 \end{vmatrix} \tag{0.1}$$

evaluates to

a) 0

b) $2b(b-1)$

c) $2(1-b)(1+2b)$

d) $3b(1+b)$

Theoretical solution

Let the determinant of the given matrix be Δ.

$$\Delta = \begin{vmatrix} 1+b & b & 1 \\ b & 1+b & 1 \\ 1 & 2b & 1 \end{vmatrix} \tag{0.2}$$

We use the row operation $R_2 \rightarrow R_2 - R_1$, which does not change the determinant's value.

$$\Delta = \begin{vmatrix} 1+b & b & 1 \\ b-(1+b) & 1+b-b & 1-1 \\ 1 & 2b & 1 \end{vmatrix} = \begin{vmatrix} 1+b & b & 1 \\ -1 & 1 & 0 \\ 1 & 2b & 1 \end{vmatrix} \tag{0.3}$$

Theoretical solution

Next, we apply the operation $R_3 \to R_3 - R_1$. This also does not change the determinant's value.

$$\Delta = \begin{vmatrix} 1+b & b & 1 \\ -1 & 1 & 0 \\ 1-(1+b) & 2b-b & 1-1 \end{vmatrix} = \begin{vmatrix} 1+b & b & 1 \\ -1 & 1 & 0 \\ -b & b & 0 \end{vmatrix} \tag{0.4}$$

The third column now has two zero entries, which simplifies the calculation.

Theoretical solution

We can now evaluate the determinant by expanding along the third column:

$$\Delta = 1 \begin{vmatrix} -1 & 1 \\ -b & b \end{vmatrix} - 0 + 0 \tag{0.5}$$

$$= ((-1)(b) - (1)(-b)) \tag{0.6}$$

$$= -b + b = 0 \tag{0.7}$$

The determinant is 0, which corresponds to option (a).